

Fingerprint Dataset Dossier

Enhancement of fingerprints degraded by occupational wear, ageing and skin damage Master's thesis · Astitva Srivastava · compiled 10 September 2026

1. Purpose and how to read this document

This document is a complete account of the image data available to the thesis. For every dataset it records: where it comes from, how it is obtained, what it actually contains as measured on disk, what its quality is, what is wrong with it, who has used it in the published literature, and what role it plays here.

Three things distinguish it from a dataset list:

- Every quantity was measured, not quoted.** Image counts, dimensions, colour modes and byte sizes come from per-image manifests generated by `scripts/make_manifest.py` and summarised by `scripts/dataset_dossier_stats.py`. Where our measurement disagrees with the publisher's specification, both are given and the disagreement is flagged. Two such disagreements were found and are material — see §4.1 and §4.6.
- Failures are reported as fully as successes.** Section 5 lists what could not be obtained and why, because the shape of the thesis is determined at least as much by the missing data as by the present data.
- Suitability is argued, not asserted.** Section 7 states plainly what this corpus does and does not license us to claim.

Live access status is tracked in `docs/datasets/REGISTRY.md`; this dossier is the point-in-time narrative version of that table.

2. Scoreboard

	Datasets	Images	On disk
Acquired and verified	11	94,243	3.0 GB
Pending — paperwork or registration in progress	8	—	—
Permanently unavailable	5	—	—

Acquired: 94,243 images. 66,267 real and synthetic fingerprints across nine manifested public datasets (1,763 MB), plus 22,336 images we generated ourselves with Anguli (654 MB), plus 5,640 non-fingerprint texture images used as degradation backgrounds (633 MB).

The headline judgement: **the acquired corpus is sufficient to build, train and evaluate the complete thesis pipeline without waiting on a single licence.** What it does not contain is expert-marked minutiae ground truth or a real occupationally-worn population; §7 sets out precisely what that costs us and how each gap is covered.

3. Acquired datasets — summary table

All figures measured from the committed manifests in `docs/datasets/manifests/`.

Dataset	Images	MB	Image size	Declared dpi	Subjects	Type
SOCOFing (+ Altered)	55,270	810.6	96×103 (99.2%)	500 (disputed)	600	Real + synthetic damage
L3-SF	8,140	569.3	320×240 / 512×512	1200 (partial)	740 ids	Synthetic, level-3
Anguli corpus (ours)	22,336	654	275×400	500	5,584 fingers	Synthetic, paired
DTD textures	5,640	633	variable	n/a	—	Backgrounds (not prints)
MINEX validation	801	114.4	167×325 – 406×662	500	54	Real, labelled
FVC2004 B (4 DBs)	320	54.7	288×384 – 640×480	500 / 512	40 fingers	Real, difficult
FVC2002 B (4 DBs)	320	41.1	300×300 – 296×560	500 / 569	40 fingers	Real, difficult
FVC2000 B (4 DBs)	320	38.1	240×320 – 448×478	500	40 fingers	Real
Neurotech U.are.U	520	62.2	326×357	512	65 fingers	Real
Neurotech CrossMatch	408	39.3	504×480	500	51 fingers	Real
FVS	168	33.0	256×256	unstated	21 fingers	Real

4. Acquired datasets — detail

4.1 SOCOFing — Sokoto Coventry Fingerprint Dataset ★ primary damage benchmark

- **Link:** <https://www.kaggle.com/datasets/ruizgara/socofing> — paper at <https://arxiv.org/abs/1807.10609>
- **Reference:** Shehu, Ruiz-Garcia, Palade & James, *Sokoto Coventry Fingerprint Dataset*, arXiv:1807.10609, 2018
- **Licence:** CC BY 4.0. Free, instant, no registration. **Redistribution permitted with attribution** — the only acquired real-subject set for which this is true.
- **Status:** downloaded, manifested, 10 September 2026.

What it is. 6,000 real fingerprints from 600 African subjects — 10 fingers each, one impression per finger — plus 49,270 synthetically altered versions of those same prints. Alterations are of three kinds, each at three severities:

Alteration types	Easy	Medium	Hard	Total
Obliteration (Obl), Z-cut (Zcut), central rotation (CR)	17,931	17,067	14,272	49,270

Verified contents (measured). 55,270 images, 810.6 MB. 600 subjects. Almost perfectly balanced across the ten finger positions (5,334–5,790 images each). 54,837 images are 96×103 px; 433 are 241×298. The 49,270 altered images are 8-bit greyscale; the 6,000 real images are stored RGBA (5,956) or RGB (44) despite greyscale content — **loaders must convert**.

Why it matters here — this is the most valuable dataset we hold. It is the only acquired set that is simultaneously (a) real fingerprints, (b) paired — every altered image has its undamaged mate, (c) labelled by damage *type*, (d) labelled by damage *severity*, and (e) labelled by *gender*. That combination supports the thesis's central experiment directly: enhance a damaged print, match it against its own clean mate, and report the change in equal error rate stratified by severity and damage type. No licence, no waiting.

Quality assessment and caveats — read before using.

- **The declared 500 dpi is not credible and we do not accept it.** At 96×103 px, 500 dpi would describe a finger 4.9×5.2 mm across. The true sampling is nearer 200 dpi. Every 500-dpi-trained model (NFIQ 2, `mindtct`, `pyfing`) assumes 500 dpi and will mis-estimate ridge frequency on these images.
Consequence: images must be rescaled to a common effective resolution before use, and NFIQ 2 scores must never be compared between SOCOFing and a genuinely 500-dpi set. This is recorded as a hard rule, not a footnote.
- **The alterations are synthetic**, generated by the authors' own software. They model obliteration, cuts and rotation — they are *not* a model of occupational wear, ageing or dryness. SOCOFing therefore measures robustness to damage; it does not stand in for the target population.
- **Gender is imbalanced roughly 4:1** (44,203 male / 11,067 female images). Sufficient for a reported stratification, too skewed for a strong claim of demographic fairness.
- One impression per real finger, so genuine pairs exist only between a print and its altered versions — not between two independent captures.

Use in the literature. Predominantly damage/alteration *classification* work — finger and gender classification, alteration-type recognition. Its use as an *enhancement* benchmark is uncommon, which is an opportunity rather than a warning.

4.2 FVC2000 / FVC2002 / FVC2004 — "B" subsets ★ the field's common currency

- **Links:** <http://bias.csr.unibo.it/fvc2000/download.asp> · <http://bias.csr.unibo.it/fvc2002/download.asp> · <http://bias.csr.unibo.it/fvc2004/download.asp>
- **Reference:** Maio, Maltoni, Cappelli, Wayman & Jain, FVC2000 / FVC2002 / FVC2004 competition reports
- **Licence:** "B" subsets free for research. The larger "A" subsets ship only with the *Handbook of Fingerprint Recognition*.
- **Status:** all twelve databases downloaded and manifested, 10 September 2026.

What it is. Three biennial competitions, four sensor databases each. Each "B" subset is 10 fingers × 8 impressions = 80 images. We hold all twelve: **960 images, 133.9 MB**. DB1–DB3 are optical, capacitive and thermal-sweep sensors respectively; DB4 is synthetic (SFinGe). FVC2002 and FVC2004 deliberately instructed

subjects to produce *difficult* impressions — exaggerated distortion, and deliberately wet or dry fingers.

Verified contents. 100% 8-bit greyscale. Four distinct image sizes per year, matching the published specification exactly in every case. Declared dpi is 500 except FVC2002 DB3 (569) and FVC2004 DB3 (512).

Why it matters here. FVC2004 DB1 is the single most directly relevant real dataset we hold: *deliberately dry and distorted fingers* are the closest freely available analogue to occupational wear. It is our Phase 0 smoke-test set and a headline evaluation set. Just as importantly, FVC is the most-reported benchmark in the entire enhancement literature, so results on it are comparable — in kind if not in exact subset — with almost every paper we cite.

Quality assessment and caveats. Genuinely low-quality real data, which is the point. The limitation is size: 80 images per database gives $10 \times C(8,2) = 280$ genuine pairs and 2,880 impostor pairs per database — enough for an EER estimate, but with wide confidence intervals, so **EER must be reported with bootstrap intervals rather than as a point estimate**. Note also that Cappelli's 2026 state of the art trained on the *A* subsets of FVC2002 DB2/DB3 and FVC2004 DB1, which we do not hold; we therefore cannot reproduce his training set exactly, and say so.

Use in the literature. The canonical benchmark of the classical era and much of the deep-learning era: Hong-Wan-Jain evaluation, FingerNet, Cappelli 2023 / 2024a / 2024b / 2026 (training and evaluation), and essentially all quality and segmentation studies.

4.3 Anguli-generated paired corpus ★ our training set

- **Link:** <https://dsl.cds.iisc.ac.in/projects/Anguli/> (generator) — corpus generated locally
- **Reference:** Ansari, *Generation of Synthetic Fingerprint Databases*, MSc thesis, IISc Bangalore, 2011
- **Licence:** Generator free for research. **The corpus is ours** — we generated it, so it carries no redistribution restriction and can be released with the thesis.
- **Status:** generated and reconciled, 10 September 2026.

What it is. 5,584 synthetic fingers, each with one clean master and three simulated impressions: **22,336 images, 654 MB**, at 275×400 px. Provenance recorded in `data/processed/anguli_dev_5584/generation.json`.

Why it matters here. This is the only source of *perfectly paired* clean/degraded data we have, and pairing is what supervised enhancement training requires. The master image is the reconstruction target; our own degradation code produces the input. Because the master is a clean synthetic ridge pattern, minutiae extracted from it serve as **pseudo ground truth**, which is how we obtain minutiae precision and recall without owning an expert-annotated dataset (see §7.2).

Quality assessment and caveats.

- Honest disclosure: the run was **requested at 20,000 fingers and interrupted at 5,584** when the machine exhausted memory. The directory is named `anguli_dev_5584` rather than `anguli_20k` precisely so that no reader mistakes its size. One torn record was detected and pruned by `scripts/reconcile_anguli.py`.
- 5,584 is nonetheless ample. Cappelli's 2026 state of the art was trained on **360 images** and outperforms FingerGAN trained on 130,000. Corpus size is not this field's bottleneck.

- Synthetic prints are not real prints. Anguli's impression variation models acquisition noise, not physiology. **All degradation in this thesis is applied by our own code** (`src/fpe/degradation/`), never by Anguli's built-in artefact flags — so that what was applied to each image is fully inspectable rather than hidden inside a closed binary. That division of labour is a deliberate methodological choice, recorded in ADR 0003.
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4.4 DTD — Describable Textures Dataset (backgrounds, not fingerprints)

- **Link:** <https://www.robots.ox.ac.uk/~vgg/data/dtd/>
- **Reference:** Cimpoi, Maji, Kokkinos, Mohamed & Vedaldi, *Describing Textures in the Wild*, CVPR 2014
- **Licence:** Free for research use.
- **Status:** downloaded — 5,640 images, 633 MB.

What it is and why it is here. 5,640 texture photographs in 47 categories. Not fingerprint data — it is the surface-background bank used to composite realistic latent-style degradation, following the SFP method. It exists to build the **control arm** of the thesis: a conventional latent-degradation model against which our wear model is compared.

Quality note that matters for the thesis's credibility. Cappelli's principal methodological criticism of DNUNets and FingerGAN is that they composited training latents onto backgrounds *extracted from the test images*, leaking test information into training. We answer that structurally rather than procedurally: DTD images were never fingerprints, so no fingerprint test image can leak through them, and DTD is partitioned **by category** via a filename hash into 35 / 8 / 4 categories = 4,200 / 960 / 480 textures. A training image cannot receive a texture from the same family as a test image, and `split="test"` cannot return a training texture. Verified disjoint at both category and file level.

4.5 MINEX validation imagery

- **Link:** <https://github.com/usnistgov/minex>
- **Licence:** NIST — public domain, permissive. Redistributable.
- **Status:** downloaded and manifested — **801 images, 114.4 MB, 54 subjects.**

What it is. The validation image set shipped with NIST's MINEX III template-conformance programme. Files are headerless raw `.gray`.

A finding of our own. The images ship without metadata, but the validation driver's C++ header carries, for all 801 images, the **width, height, quality band and finger position**. We parse it (`src/fpe/data/minex.py`), which yields a ready-made stratification axis — 8 finger positions, no thumbs, 54 subjects, **132 images labelled poor or fair** — on permissively licensed data. Every image has near-unique dimensions (788 distinct sizes across 801 images), as they are individually cropped.

Role here. A secondary real-data evaluation set whose particular value is the finger-position and quality-band labels, which allow stratified reporting that most published work omits.

4.6 L3-SF — Level-3 Synthetic Fingerprints

- **Link:** <https://github.com/luannd/L3-SF>
- **Reference:** Wyzykowski, Segundo & Lemes, *Level Three Synthetic Fingerprint Generation*, ICPR 2020
- **Licence:** CC BY-NC-SA 4.0 — non-commercial, share-alike.
- **Status:** downloaded and manifested — **8,140 images, 569.3 MB.**

What it is. Synthetic fingerprints carrying level-3 detail — sweat pores and scratches. 740 identities, with a pore-annotated subset carrying TSV pore coordinates.

A correction to the published description. The dataset is advertised as 1200 dpi. Our measurement shows the 1200 dpi figure applies to **only 740 of the 8,140 images** — the 512×512 pore-annotated subset. The remaining 7,400 are 320×240, which at 1200 dpi would describe a finger 6.8×5.1 mm across and is therefore impossible. The main set's true resolution is recorded as **unknown, to be measured from ridge period**, rather than guessed. All images are stored RGB despite greyscale content.

Role here. Secondary. Level-3 pore detail is beyond the 500-dpi civil-capture regime this thesis targets, so L3-SF is held for a possible ridge-continuity analysis rather than for training.

4.7 Neurotechnology sample databases

- **Link:** <https://www.neurotechnology.com/download.html#databases>
- **Licence:** Free sample data provided by the vendor for evaluation.
- **Status:** both downloaded and manifested.

Subset	Images	MB	Size	dpi	Structure
CrossMatch Verifier 300	408	39.3	504×480	500	51 fingers × 8
U.are.U 4000	520	62.2	326×357	512	65 fingers × 8

Role and quality. Clean, consistent, single-sensor optical capture with eight impressions per finger. Their value is precisely that they are *good* data: eight impressions per finger gives $51 \times C(8,2) = 1,428$ and $65 \times C(8,2) = 1,820$ genuine pairs — by some margin the largest genuine-pair counts among our real multi-impression sets, and therefore the tightest EER estimates. They serve as the **clean control condition**: a method that damages good images is disqualified regardless of how it performs on bad ones.

4.8 FVS — Fingerprint Verification System sample database

- **Link:** <http://fvs.sourceforge.net/>
 - **Licence:** Open-source project sample data.
 - **Status:** downloaded and manifested — 168 images, 33.0 MB, 256×256, 21 fingers × 8.
 - **Quality:** small, dpi unstated, stored RGB despite greyscale content. A minor supplementary set only.
 - **Acquisition note:** the host's TLS certificate has expired. Python's `urllib` refuses the connection; `curl` accepts it. Recorded in the checksum log so the download remains reproducible.
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5. Not acquired — and why

Reporting these fully is not an admission of incompleteness. The availability situation in this field is itself a documented research problem (§6), and the thesis is deliberately designed so that none of these is load-bearing.

5.1 Blocked by a broken provider — ChaLearn LAP Track 3

- **Link:** <https://chalearnlap.cvc.uab.cat/dataset/32/description/>
- **What it was to be:** 84,000 paired clean/degraded images — the intended primary training set, and the training set of a substantial part of the denoising literature (Mansar; FPD-M-Net).
- **What happened:** the registration system is broken. Sign-up returns HTTP 500; password reset and repeated sign-up across several email addresses all fail. **Assessed as unrecoverable.**
- **Consequence and response:** ADR 0003. We replaced it in full. Anguli supplies the clean masters, and we reimplemented ChaLearn's degradation ourselves. ChaLearn applies **nine** artefact types — blur, brightness, contrast, elastic transform, occlusion, scratches, resolution reduction, rotation, and compositing onto background textures — and all nine are now implemented in `src/fpe/degradation/latent.py`, seeded, with a per-image record of which artefact fired with which parameters. This is arguably a better position than holding the original, because what was applied to each image is inspectable. We do not claim exact comparability: ChaLearn published its artefact *types* but not their parameter distributions.

5.2 Awaiting registration or approval

Dataset	Link	Why we want it	Blocker
CASIA-FingerprintV5	http://biometrics.idealtest.org (id 7)	500 subjects $\times$ 8 fingers $\times$ 5 impressions. The population explicitly includes workers and waiters , and difficult impressions were requested — the closest free analogue to an occupationally worn population.	Registration plus manual human approval. Confirmed live via the portal API on 10 Sep 2026. Walkthrough written in <code>docs/datasets/casia-access.md</code> .
CASIA Fingerprint Subject Ageing	http://biometrics.idealtest.org (id 15)	3,664 images. Ageing is a named degradation cause in this thesis's title , and public longitudinal fingerprint data is close to non-existent.	Same registration route. Found by us via the portal API; absent from both the thesis plan and the literature review until now. Longitudinal structure unconfirmed until the accompanying note is readable.
PrintsGAN	MSU — Engelsma, Grosz & Jain	525,000 images, 35,000 fingers, multiple impressions per identity — synthetic pretraining at scale.	Signed agreement sent to Steven Grosz, MSU. Not paperwork-free.
Tsinghua SD27 annotation package	http://ivg.au.tsinghua.edu.cn/dataset/NIST.php	Manually labelled orientation fields, period maps, skeletons and ROIs — the class of ground-truth supervision Cappelli's orientation and frequency networks were trained on.	Request not yet sent. Note that these annotate SD27 images, which are themselves withdrawn (§5.4).
FVC manual segmentation ground truth	Thai, Huckemann & Gottschlich, <i>PLOS ONE</i> , 2016	Hand-marked segmentation masks for FVC images — what Cappelli trained his segmentation on.	Request not yet sent.

5.3 Licensed — signature required, long lead time

All route through the IIIT-Delhi Image Analysis and Biometrics lab (databases@iab-rubric.org) or NIST, and all require an institutional signature.

Dataset	Route	Why we want it	Status
Rural Indian Fingerprint DB ★	databases@iab-rubric.org	The target population of this thesis — rural and urban Indian subjects with genuine occupational degradation. The benchmark of the Joshi research line.	Not started. Requires a Registrar / Head-of-Institution signature . Request drafted in docs/correspondence/ .
MUST	Same route	~21,000 impressions with manually marked minutiae , PPI and semantic masks. This is the dataset that would give us true minutiae F1 (§7.2).	Not started; drafted.
IIIT-D MOLF	Same route	19,200 images, 100 Indian subjects, five capture methods. Used by the Joshi line and MLFGNet.	Not started; drafted.
NIST SD302 (N2N) and SD300	https://nigos.nist.gov/datasets/sd302/request	SD302: multi-sensor real capture, 200 subjects × 10 fingers. SD300: 888 subjects of naturally poor, uncooperative ink-card captures.	Not started; institutional email and manual NIST review required.

Assessment. These are the highest-value outstanding items, and the Rural Indian DB is the single most important. They are also the slowest. The thesis is therefore structured so that **no phase blocks on any of them**: the Rural Indian DB is a *confirmatory* external test set. If it arrives it becomes the headline external validation; if it does not, every claim in the thesis still stands on data we already hold.

5.4 Permanently unavailable

Dataset	Status	Consequence for this thesis
NIST SD27	Withdrawn January 2017	258 real crime-scene latents with expert-validated minutiae. This is the principal benchmark of the entire latent-enhancement literature — <i>including the 2026 state of the art</i> . Published numbers on it cannot be reproduced by anyone not already holding a legacy copy.
NIST SD4 / SD9 / SD10	Discontinued	Historical baselines unreproducible.
NIST SD14	Uncertain; treated as unavailable	Was the 27,000–30,000-image training source for DNUNets, FingerGAN and SFP, so those training sets cannot be reconstructed either.

6. The availability crisis — why this matters scientifically

This is worth stating explicitly, because it converts an obstacle into a contribution.

The field's principal benchmark has been unavailable for **nine years**. SD27 was withdrawn in January 2017 for lacking the documentation NIST requires for distribution. Yet it remains the evaluation set in papers published in 2024, 2025 and 2026 — including the current state of the art — because established groups retain legacy copies.

The consequences are concrete: new entrants cannot reproduce published numbers, comparisons are not independently verifiable, and the reported ranking of methods rests on a dataset no reviewer can check. The field has begun to respond. The ATVS group at Universidad Autónoma de Madrid built GCDB from Guardia Civil casework after finding SD27 undistributable; LFIW was constructed as a replacement; and Wang et al. (2025) moved their evaluation to MOLF and MUST.

This thesis follows that line, and the constraint becomes a design principle: a reproducible, precision-first, matcher-centric protocol built entirely on **obtainable** data, released with split files and scripts. A protocol a reader can actually re-run is worth more than a number on a dataset they cannot have.

7. What this corpus does and does not support

Stated plainly, so that no claim in the thesis outruns its evidence.

7.1 Supported without further acquisition

Capability	Evidence base
Supervised training of an enhancement model on clean/degraded pairs	Anguli — 5,584 fingers × 4 images — plus our degradation code
A controlled comparison of degradation models: our wear model vs. a conventional latent model	Both implemented in <code>src/fpe/degradation/</code> ; DTD supplies leakage-free backgrounds
Matcher EER before vs. after enhancement, on real prints	FVC ×12, Neurotech ×2, FVS, MINEX — all multi-impression
EER on real damaged prints, stratified by damage type and severity	SOCOFing — 49,270 altered images with mates, 3 types × 3 severities
NFIQ 2 quality-distribution shift	All sets; NFIQ 2 installed and verified working
Stratification by finger position	MINEX (8 positions, labelled) and SOCOFing (10 positions, balanced)
Stratification by gender	SOCOFing — imbalanced 4:1, so reportable but not a fairness claim
Failure-to-acquire / failure-to-enrol reduction	Any set; measured as the rate at which the extractor returns a usable template
Leakage-free training, verifiably	DTD category-level partition; no fingerprint test image can reach training

7.2 Not supported, and how each gap is covered

Gap	Why	Cover
Expert-marked minutiae ground truth on real prints , so no true minutiae precision / recall / F1 directly comparable to Cappelli's table	Requires SD27 (withdrawn), or MUST or MOLF (licence not yet granted)	Minutiae precision and recall are computed against pseudo ground truth from clean Anguli masters and reported as such — never presented as comparable to published SD27 numbers. Real-data claims rest on matcher EER , which requires no annotation. MUST is being pursued in parallel.
A genuinely occupationally worn population	Rural Indian DB not yet licensed	CASIA-V5 (workers, waiters) in registration; FVC2004 DB1 (deliberately dry and distorted) held now. The wear model's realism is validated by measurement — NFIQ 2 distribution comparison and a domain classifier — rather than asserted.
Longitudinal ageing data	CASIA Subject Ageing awaiting approval	Ageing enters the wear model as a parameterised physiological effect. The empirical ageing claim is deferred unless the dataset arrives.
Manual orientation / frequency / segmentation ground truth	Tsinghua and Thai et al. packages not yet requested	We use Cappelli's <i>pretrained</i> networks via <code>pyfing</code> for orientation, frequency and segmentation rather than retraining them, inheriting his supervision without needing his labels.
Direct numerical comparison with the 2026 state of the art	SNFEN's published numbers are on SD27, which we cannot obtain	We run <code>pyfing</code> 's released GBFEN and SNFEN <i>ourselves</i> , on <i>our</i> benchmark. A comparison we execute on obtainable data is stronger than a number copied from a table on a dataset nobody can check.

8. Tooling status

Tool	Purpose	Status
NFIQ 2	ISO/IEC 29794-4 quality score, 0–100	Installed and verified — <code>tools/bin/nfiq2.exe</code> , scores FVC2004 DB1_B correctly
Anguli	Synthetic generator	Working — Windows binary and Qt source both held
pyfing	Cappelli's segmentation / orientation / frequency / GBFEN / SNFEN — strongest baseline and best starting codebase	Not yet installed. Verified to be Keras 3 , which means it can run on the PyTorch backend, so the TensorFlow-on-Windows GPU problem is avoidable. Also ships LEADER, an end-to-end minutiae extractor.
NBIS (<code>mindtct</code> , <code>bozorth3</code>)	Free minutiae extractor and matcher — baseline matcher #1	Not installed. No C compiler present on this machine; a toolchain must be provisioned to E:.
SourceAFIS	Independent matcher #2 — two independent matchers is a protocol requirement	Not installed
MCC (Minutia Cylinder-Code)	Optional third matcher; Bologna, academic use	Not installed

9. Provenance and reproducibility

- Every acquired dataset has a per-image manifest under `docs/datasets/manifests/` recording relative path, **SHA-256**, byte size, pixel dimensions, declared dpi and colour mode. Manifests are committed; **images never are**.
- The per-dataset figures in this document are produced by `scripts/dataset_dossier_stats.py` from those manifests into `docs/datasets/manifest_summary.csv`. Re-running the script regenerates them.
- Downloads are performed by `scripts/download_group_a.py` and `scripts/download_group_b.py`, which verify checksums, cache to avoid re-fetching, and refuse to overwrite existing files.
- Live access status is maintained in `docs/datasets/REGISTRY.md`.
- **Licence discipline.** IAB, NIST, CASIA, FVC-A and LivDet data may not be redistributed. `.gitignore` blocks image formats by default and is never overridden for dataset images. SOCOFing (CC BY 4.0), MINEX (public domain) and our own Anguli corpus are the only holdings that could be redistributed, and only SOCOFing and the Anguli corpus are planned for release alongside the thesis.

Compiled 10 September 2026. All figures measured from committed manifests; access status per `docs/datasets/REGISTRY.md`.